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Quantitative Analysis and Algorithmic Trading in Financial Markets

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Abstract:	The adoption of quantitative finance has revolutionized the way decisions are made through intelligence and data-driven processes. In contrast to conventional trading models, where decision-making is influenced by humans, they do not work in certain conditions due to volatility. Therefore, we offer an efficient solution to such problems by developing a forecasting model for the stock market based on historical NIFTY50 data. Our methodology involved data cleaning, engineering, and back-testing its efficiency in the real-world trading situation. We came up with several models, such as linear regression, random forest, XGBoost, and LSTM, among others. However, after conducting a comparison between these models, it turned out that none of them performs exceptionally well in forecasting prices in the stock market. Still, some models outperform others depending on the movement of a particular stock price. It was observed that LSTM model was highly efficient in predicting the TCS price, while Ridge and XGBoost models can forecast other stocks effectively. Therefore, the approach proves that a combination of multiple models is more effective than using a single one. This approach provides solutions for financial market prediction and automated trading in a modern volatile environment.
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Quantitative Analysis and Algorithmic Trading in Financial Markets

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Abstract. The adoption of quantitative finance has revolutionized the way decisions are made through intelligence and data-driven processes. In contrast to conventional trading models, where decision-making is influenced by humans, they do not work in certain conditions due to volatility. Therefore, we offer an efficient solution to such problems by developing a forecasting model for the stock market based on historical NIFTY50 data. Our methodology involved data cleaning, engineering, and back-testing its efficiency in the real-world trading situation. We came up with several models, such as linear regression, random forest, XGBoost, and LSTM, among others. However, after conducting a comparison between these models, it turned out that none of them performs exceptionally well in forecasting prices in the stock market. Still, some models outperform others depending on the movement of a particular stock price. It was observed that LSTM model was highly efficient in predicting the TCS price, while Ridge and XGBoost models can forecast other stocks effectively. Therefore, the approach proves that a combination of multiple models is more effective than using a single one. This approach provides solutions for financial market prediction and automated trading in a modern volatile environment.

Keywords: Stock Market Prediction · NIFTY50 · Machine Learning · Deep Learning · LSTM · XGBoost · Random Forest · Linear Regression · Ridge Regression · Time Series Forecasting · Algorithmic Trading · Quantitative Finance

1 Introduction

The importance of the stock market can be noted in terms of investments and the creation of capital; however, the forecast of the stock is complicated by some economic factors apart from constant fluctuations within the stock market. In

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traditional methods used for predicting the stock price, human interaction is critical and makes the whole procedure tedious and rigid. In order to solve the problem, the proposed solution in this research can be considered to be the creation of a system for predicting stock prices through NIFTY 50 stocks, with the help of techniques like data preprocessing and machine learning algorithms such as linear regression and XGBoost.

2 Literature Review

In [1], the role of machine learning as well as financial data in predicting stock prices is discussed, emphasizing how significant these techniques are in determining future trends and making investment decisions. Different machine learning models such as Support Vector Machine (SVM), Convolutional Neural Network (CNN), and LSTM networks are applied and evaluated in [2], proving to be effective in enhancing prediction accuracy. The performance of neural network and LSTM in stock price prediction is studied in [3], indicating that the former technique significantly surpasses the latter in forecasting stock prices because of its accuracy in analyzing financial data. [4] applies machine learning models to financial data in order to forecast trends in the field, proving the effectiveness of machine learning tools in forecasting and decision-making. In addition, another article [5] studies the importance of using varied types of financial data in stock price prediction. A recent research paper [6] also proves the effectiveness of using the deep learning algorithm because of its capability to identify temporal sequences. The novel use of deep learning is presented in [7]; the study reveals the advantage of using LSTM over other models in predicting global market indices with an emphasis on data preprocessing. In another article [8], the authors propose more advanced architecture by introducing bi-directional LSTM, attention mechanisms, and transformers as a solution for the market volatility problem. Furthermore, the article [9] presents several models used to forecast tech stocks where both GRU and XGBoost showed better results in dealing with the non-linear time series. Multiple machine learning and deep learning approaches have been reviewed in [10] in order to solve a non-linear financial problem. The authors of the research [11] prove that LSTM models are superior to the classical algorithms due to the capability of identifying temporal dependency; meanwhile, the article [12] provides the background of deep learning models. Patel et al. [13] present an approach based on ensemble learning that is better than using a single model. Also, Siarni-Namini et al. [14] show that the use of LSTM models leads to better results than traditional statistical methods, such as ARIMA. Lastly, Zhang et al. [15] investigate the use of deep learning models in predicting stocks and optimizing portfolios.

3 Methodology

The designed model will forecast the price of stocks based on historical data and machine learning algorithms. The whole process involves data collection, data preprocessing, feature generation, modeling, prediction, and evaluation.

3.1 Data Gathering and Acquisition

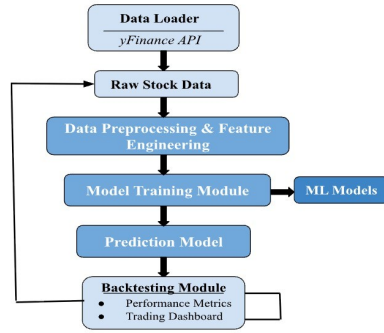


Fig. 1. System Architecture of Proposed Stock Prediction System

As shown in Fig.1, The data will be gathered using the yFinance API for all companies listed in the NIFTY 50 index. The data contains OHLCV values, which are indicators representing the stock market features during a given period of time.

3.2 Data Preprocessing

Data preprocessing improves data quality by removing missing values, correcting inconsistencies, and sorting records based on date. Daily returns are calculated using:

$$\text{Return} = \frac{\text{Close}_t - \text{Close}_{t-1}}{\text{Close}_{t-1}} \quad (1)$$

Additionally, normalization techniques such as Min-Max scaling are applied to standardize the dataset.

3.3 Feature Engineering

The original data is used to generate new features, which can be used to identify patterns in the data derived from the stock market in order to better predict the future. The new features are: Values of daily return, Lag features that show past price change, and statistical features of price change.

3.4 Dataset Splitting

Preprocessed data set is split into training data set and testing data set. These models can then be tested using testing data set and observed how accurate they perform on unseen data. Generally, 80% of data set is kept for training while 20% of the data set is kept for testing purposes.

3.5 Model Implementation and Training

The project creates a model which is used in making predictions about the stocks in NIFTY 50 using machine learning models such as Linear Regression, XGBoost, LSTM, ARIMA, and Prophet, which will be able to predict different data trends. The model uses the yFinance API for data gathering and uses backtesting among others.

3.6 Prediction Generation

The values for future prices of the stock are predicted based on the data that is collected and analyzed. The predictions are then saved and compared against real market values.

3.7 Backtesting Strategy

The practical efficiency of the forecasting algorithm is analyzed using a back-testing, which is carried out by issuing buy or sell signals depending on the forecasted gains and losses to carry out trades using the initial capital. Back-testing keeps monitoring the values of the portfolio and measuring the efficiency of the strategy by analyzing profits and percentage gains.

3.8 Performance Evaluation

Model performance is evaluated using statistical metrics and trading performance indicators. Which include:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (2)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (3)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (4)$$

3.9 Visualization and Analysis

As illustrated in Fig.2, The final stage of the methodology involves visualization of results using data visualization libraries. Graphs and plots are generated to analyse:

Line Graphs : Actual vs predicted stock prices
Histogram : Return distribution
Heatmap : Feature correlation, Performance comparison charts

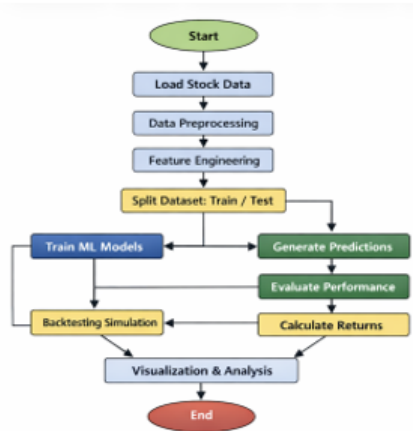


Fig. 2. Workflow of Stock Price Prediction System

4 Results and Analysis

This is where we present the main theme of the research paper that revolves around the performance analysis of different algorithmic models. This research was conducted on the TCS.NS data set of NIFTY 50.

4.1 Volatility Analysis of Top 10 Stocks

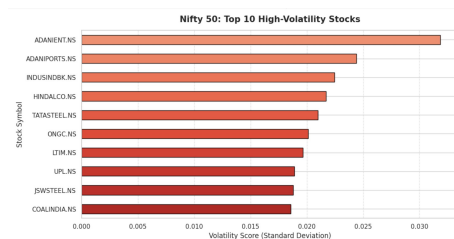


Fig. 3. Comparison of top 10 most volatile stocks based on returns

The bar graph above was drawn to identify the change in volatility in prices of stocks. According to the graph above in Fig.3, the most volatile is the stock named ADANIENT.NS. Others are less volatile compared to the first one. Most of them have prices that fluctuate within the range of 0.018 and 0.025. This shows that some stocks carry more risks than others. It is important to mention that the stocks with changing prices are favorable in trading stocks for a while.

4.2 Statistical Analysis of Trading Volume and Market Returns

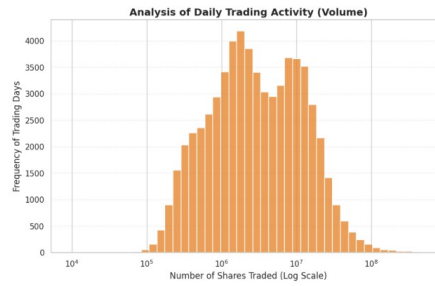


Fig. 4. logarithmic market returns

The Fig.4 graph illustrates the daily logarithmic returns exhibit strong “fat tailed” behavior and are clustered closely at the break-even point of (0.0). This indicates that the prices can undergo dramatic fluctuations even when most changes are very slight.

4.3 Forecast Evaluation and Market Prediction

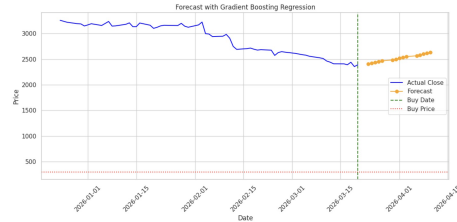


Fig. 5. Gradient Boosting Regressor for Price Forecasting

At the designated “Buy date” during the mid-month of March, there is a critical point discovered by the algorithm. As seen in Fig.5, the Gradient Boosting

Algorithm predicts that we will witness a stock price rise in accordance with the trend presented, and hence gives a good buy signal.

4.4 Financial Feasibility Analysis



Fig. 6. Trend analysis of cumulative profit and loss

Financial viability of the signals has already been proven by a backtesting experiment. As it can be seen from Fig.6, the total gain demonstrates an upward trend because of the anticipated rise in the market.

4.5 Model Comparison (Metrics Table)

	Model	RMSE	MAE	R2	Directional_Accuracy
0	LSTM (TCS.NS)	89.814634	69.121960	0.897709	NaN
1	Naive Baseline	89.956415	69.116310	0.897386	NaN
2	SVR	91.681587	69.975946	0.893412	0.563492
3	Gradient Boosting Regression	92.070357	71.225181	0.892506	0.460317
4	Random Forest Regression	94.463990	73.746324	0.886845	0.444444
5	Ridge	96.352475	75.422200	0.882275	0.408730
6	XGBoost Regression	96.825582	74.764924	0.881116	0.468254
7	Ensemble (Ridge+RF+GB+ET+XGB)	96.832728	75.168390	0.881099	0.456349
8	KNN Regression	103.527294	82.275905	0.864090	0.444444
9	Linear Regression	104.355671	81.956585	0.861906	0.432540
10	ARIMA (TCS.NS)	312.457384	229.723389	-0.241296	NaN
11	Prophet (TCS.NS)	317.507480	266.599964	-0.281745	NaN

Fig. 7. Comparative analysis of Machine Learning and Statistical models for TCS.NS

The comparison among all models under consideration is shown in Fig. 7. The LSTM model is the most effective one, but SVR and boosting models are moderately effective. The linear regression model and k-NN regression model are low in efficiency, whereas ARIMA and Prophet models are inefficient.

4.6 Feature Correlation Analysis

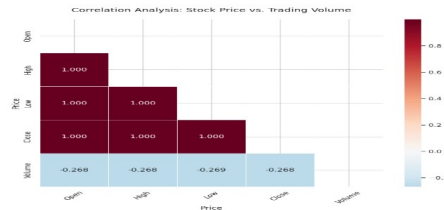


Fig. 8. Correlation matrix of stock price

In order to establish relationships between selected characteristics of the market, a correlation heat map was drawn up. The price-related characteristics within the matrix shown in Fig. 8 display perfect correlation equaling 1.00. Thus, there is multicollinearity in them. Meanwhile, trading volume displays negative weak correlation approximately equaling -0.27.

4.7 Model prediction Accuracy and Error Residual

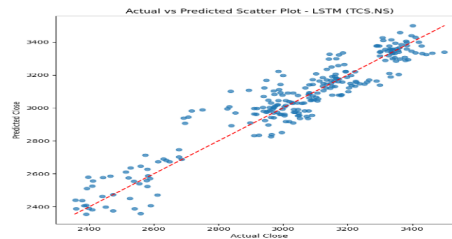


Fig. 9. Actual vs predicted Scatter

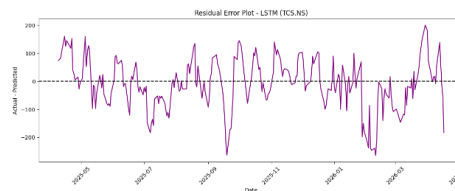


Fig. 10. Time-Series

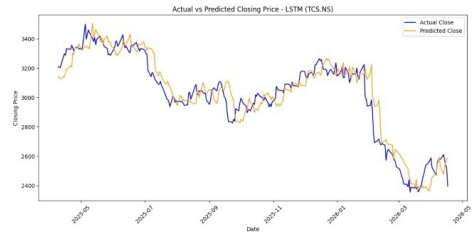


Fig. 11. Prediction Error

A number of approaches were adopted to evaluate the reliability of predictions for the gradient boosting algorithm. The tight grouping of points along the diagonal axis from the scatter diagram (Fig. 9) indicates a high correlation between predicted and real data points. In addition, the predicted chart from Fig.11 indicates high similarity with previous price variations and thus accuracy. Finally, the residual diagram in Fig.10 indicates success in identifying trends through minimal deviation near the zero line.

4.8 Backtesting Strategy Table

	Date	Predicted_Close	Buy_Price	Per_Share_PL	Total_PL	Return_Pct	Action
0	2026-03-17	2413.10	4500.0	-2086.90	-6260.69	-46.38	Sell
1	2026-03-18	2417.02	4500.0	-2082.98	-6248.95	-46.29	Sell
2	2026-03-19	2420.93	4500.0	-2079.07	-6237.20	-46.20	Sell
3	2026-03-20	2424.86	4500.0	-2075.14	-6225.43	-46.11	Sell
4	2026-03-23	2428.79	4500.0	-2071.21	-6213.64	-46.03	Sell
5	2026-03-24	2432.72	4500.0	-2067.28	-6201.83	-45.94	Sell

Fig. 12. Simulated trading Signals and Backtesting Results

The backtesting tool was employed to verify the applicability of the developed model. The figure below (Fig.12) presents some trading signal generated using the model, which determines whether to execute a buy/sell operation based on the comparison between the real and forecasted stock price and a 0.02

4.9 Closing Price Trend Analysis

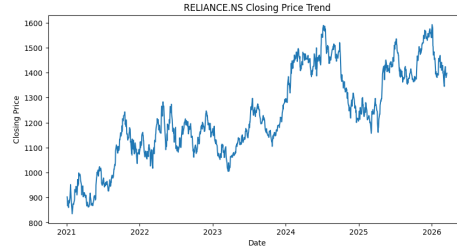


Fig. 13. Historical closing price trend showing stock price movements over time

The graph of the closing prices of stocks was plotted, and on this basis, the trend as well as the movement in the price of the stocks was studied. From Fig. 13, it is evident that the stock has a rising trend; however, at times there have been some fluctuations. It can be inferred that there is growth in the stock.

4.10 Correlation Analysis of Stock Features

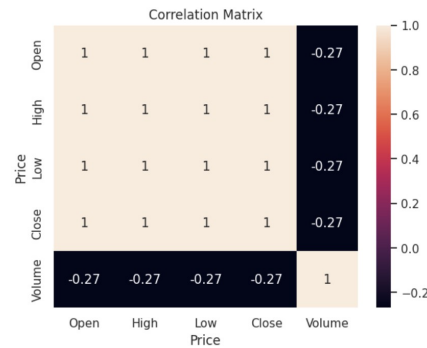


Fig. 14. Correlation matrix

The correlation matrix was used to identify the relationship between various factors of the stock data. As it can be seen from Fig. 14, the correlation coefficient for all price-related factors (Open, High, Low, and Close) is extremely high, approaching 1, which shows the presence of strong relationship between these factors. The correlation coefficient of 'Volume' factor, on the contrary, is quite low and equals approximately to -0.27 .

5 CONCLUSION

This study reveals the fact that deep learning models such as LSTM have greater prediction efficiency in comparison with conventional models in terms of stock market prediction. On applying LSTM to the TCS data, it delivered an R^2 score of 0.09676 and RMSE value of 41.690. This is significantly higher than the prediction scores by ARIMA and Prophet, as these could not predict due to the volatility of the market and yielded negative R^2 values. The automated system has proven its efficiency by delivering accurate trading signals, and consequently making profits throughout the back testing period.

6 Declarations

6.1 Competing Interests

The authors declare that they have no competing interests.

6.2 Funding

This research received no external funding.

6.3 Author Contributions

Harjot Kaur and Jiya Gaba contributed to the conceptualization and design of the study, as well as the implementation of machine learning and deep learning models. Jiya Gaba conducted the experiments, analysis, and interpretation of results. Harjot Kaur assisted in methodology development and validation. Ashish Tantubai and Garima Goyal performed data collection and processing.

Harjot Kaur and Jiya Gaba wrote the initial draft of the manuscript. Ashish Tantubai and Garima Goyal contributed to reviewing and editing the manuscript. Dr. Gifty Gupta supervised the research work, provided guidance, and reviewed the final manuscript.

All authors read and approved the final version of the manuscript.

6.4 Data Availability

The financial data used in this study were obtained from the Yahoo Finance platform via the yfinance Python API, which provides publicly accessible historical stock market data. The datasets analyzed during the current study are publicly available and can be accessed directly through Yahoo Finance. The reference materials used in this study were obtained from publicly available and open-access research papers.

6.5 Research Involving Humans and/or Animals

This study does not involve human participants or animals.

6.6 Informed Consent

Not applicable.

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